Exploring the Potential of a Centralised University Assessment Database

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Abstract— **This project investigates whether a centralised, university-built repository could improve access to educational content across the university, and what could be built on top of it. Using assessment data as a test case, we developed a prototype that extracts information from assessment briefs, validates it with a Pydantic schema, and stores it in a MySQL database covering module information, learning outcomes, assessment requirements and marking rubrics. A Retrieval-Augmented Generation (RAG) chatbot was built on top of the database.**

Keywords - centralised repository, assessment data, relational database, retrieval-augmented generation (RAG), structured data extraction, large language models (LLMs).

I. Introduction

This project investigates whether and how a centralised, university-built repository could improve access to educational content for students, lecturers and administrative staff, and what could be built on top of it.

We use assessment data as a test case. We built a prototype that extracts information from assessment briefs in PDF and Word format, validates it against a Pydantic schema, and stores it in a MySQL relational database. The database holds module information, learning outcomes, assessment requirements and marking rubrics. Two applications were built on it: a chatbot that answers student questions, and a tool that gives feedback on draft work against the marking criteria. Both use Retrieval-Augmented Generation (RAG), meaning that the AI model answers from retrieved university documents and data rather than from its general knowledge. The prototype was tested on 19 briefs from different departments.

The report addresses four questions:

1. Can assessment information from separate documents be brought into one structured database?
2. Does centralisation give a clearer overview across modules, and can it reveal wider issues in assessment briefs?
3. What applications can be built on one shared repository, such as student and staff tools, educational analytics and AI?
4. How does building a system in-house compare with buying one in, in terms of cost and security?

The prototype provides evidence for the first three questions. The fourth, along with the use of analytics to identify where students and lecturers struggle, is discussed through analysis because the prototype cannot measure it.

The report is written so that others at the university can reuse the work. It describes what was built, why each design choice was made, how the prototype performed, and the problems we met. On 19 briefs, the prototype extracted more than half the briefs correctly, and the central database supported queries that separate documents could not, such as finding clashing deadlines and missing information. The main limitations are the small sample from one department, the quality of the source documents, and the use of an external AI service.

II. Problems and Aims

A. Problems

In universities, assessment information is kept in separate documents for each department. Lecturers create a brief for each module and upload it to Moodle. This limits the access that students, lecturers and administrative staff have to information, and makes it difficult to see deadlines, weightings, assessment types and requirements across a programme, and how they relate to one another. Students, lecturers and administrative staff are all affected by data silos, which are defined as the hoarding of information within specific departments [1]. With 67% of global higher education institutions struggling with this problem, isolated data is clearly common across departments, and it hinders information sharing and leads to lower work efficiency and poorer decision-making [2].

This also directly affects progressive learning in universities. Students find it difficult to get feedback and build relationships with lecturers, especially in large cohorts, and often receive feedback only after submitting an assignment. This could be addressed by a tool built on top of a centralised assessment repository that gives draft feedback or answers questions. Students who want to see their upcoming deadlines currently have to organise them themselves, and they must also relate the content of each brief and module to one another.

Lecturers have no insight into how well students have understood a brief, or how accessible and usable their assessment is. They also cannot see which questions are asked most frequently about assessments, which could reflect on module teaching.

Administrators cannot see whether lecturers are using the assigned templates to create briefs, nor which assessment types modules use most often. Analytics on how early or late students submit their work are also unavailable..

B. Aims

Given these issues, we aim to investigate whether a centralised assessment repository will improve access to assessment information. We will also evaluate whether the data collected can be used to identify wider issues within modules, teaching or assessment briefs. For a centralised database implemented on a wider scale, it is important to ask what impact it would have.

Once the prototype is built, we will investigate what kinds of tools could be built on top of it, whether a centralised

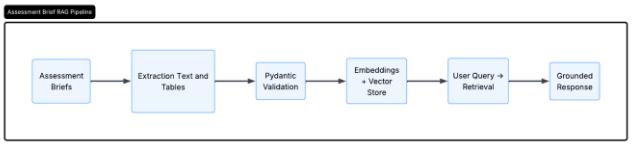
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repository would allow effective integration across departments, and whether a university-built system would help with cost and security compared with importing one from an external company. The objectives are to:

- Build a prototype that stores assessment briefs in one structured database
- Evaluate how accurately it extracts the information.
- Build tools on top for students and staff.
- Discuss cost, security and wider use.

III. Approach and Prototype

The main aim of the prototype was to create a structured database of assessment information. Information is extracted from a set of assessment briefs, validated against a Pydantic schema, and stored in a MySQL relational database. We checked the extraction by querying the database in MySQL Workbench and comparing the results with the original briefs. A chatbot was then built on top as an example application.



A. Schema

We defined the Pydantic schema before extraction to identify the key pieces of information in each brief. Starting from one brief and the standard template, we separated the information into discrete facts (module code, title, deadline, word count, weighting and submission format) and texts and lists (learning outcomes, task description, guidance and marking criteria).

The texts split further into long-form text and tables. The marking rubric, a grid of criteria and grade bands, needed its own nested class inside the marking criteria list.

Validators check values such as dates and word counts. Where a brief gives a deadline only in words, such as “Week 10”, both the parsed date and the original wording are stored. We tested the schema by checking whether any information in the brief had no field to go into, and revised it until nothing was left out.

B. Extraction

Clean text was first extracted from each file, using PyMuPDF for PDFs and python-docx for Word documents. The text was then sent to Gemini, which fills in the schema.

We chose Gemini because the university already has a partnership with it. The model is told to use only information stated in the document and to return null otherwise, and Pydantic validates its output.

We tested the pipeline on one brief and checked it field by field, then ran it on further briefs to see how well it generalised.

C. Database

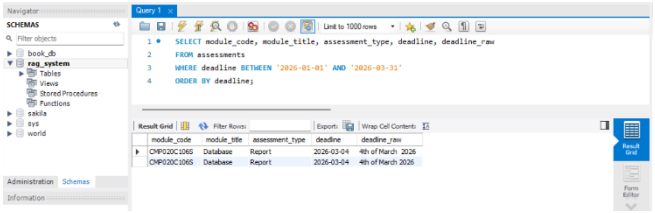
Validated records are stored in a MySQL relational database. Single-value facts sit in one assessments table. Lists (learning outcomes, checklist items, marking criteria and their grade bands) sit in linked tables so that each item can be counted and compared. Each record keeps its source file so that it can be traced back to the original document.

D. Search & Queries

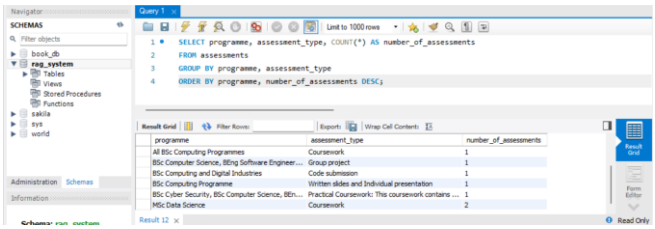
SQL queries answer cross-module questions that are hard to answer from separate documents. These include assessments within a time period, assessment types across a programme, modules with several assessments, weightings, missing information, and deadlines that are close together.

For example:

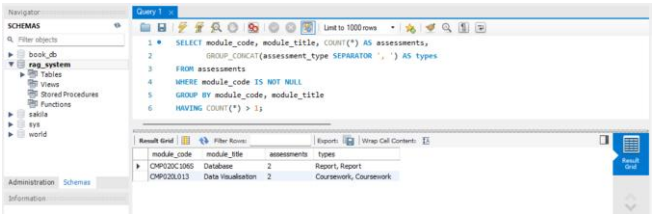
Assessments within a particular time



Assessment types across a programme



Modules with multiple assessments



E. Chatbot

The chatbot is a Streamlit app. When a student selects an assessment, the facts, criteria and learning outcomes come directly from the database. The long text sections are searched with a local embedding model (all-MiniLM-L6-v2), restricted to that assessment.

Gemini then answers using only this material, and says when the answer is not in the brief. We tested it with five questions covering facts, rules, meaning and information not in the brief, recording a pass or failure for each and the cause of each failure. We kept the number of questions small because of frequent 503 errors (the server was temporarily unable to handle a request because of overload). We also planned to compare the results with a baseline that sends the whole brief to the model.

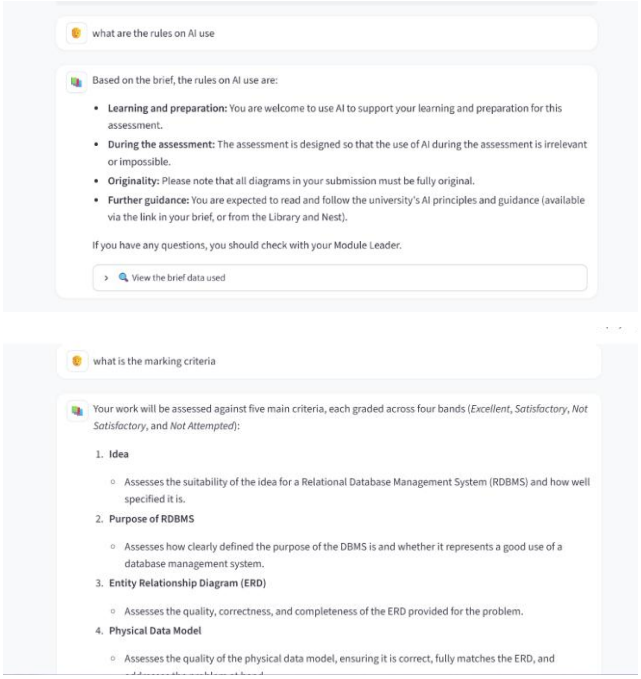
F. Feedback

For draft feedback, the system loads the selected assessment’s criteria and grade bands from the database and gives them to the model, together with the student’s draft, as a checklist.

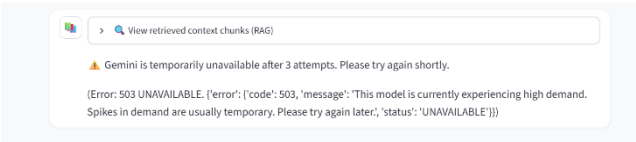
For each criterion, the model names the grade band that the draft currently matches, gives evidence from the draft, and says what would be needed to reach the next band. The word

count is calculated in code rather than by the model, and the tool gives feedback only, not a mark.

Pass example:



Failure example:



IV. Results

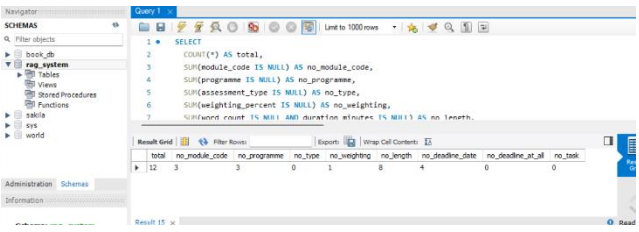
A. Ectraction Accuracy

We ran the pipeline on 19 assessment briefs (13 PDF and 6 Word) from multiple departments. Each stored record was compared field by field with the original document. We were able to successfully extract 11 documents. This equates to 57.9% extraction accuracy.

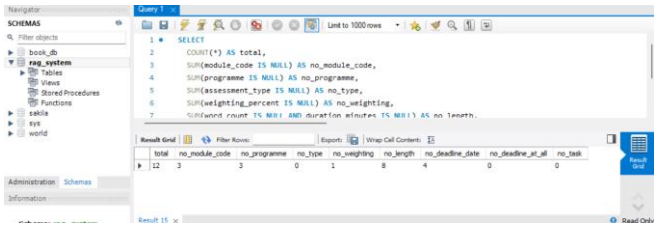
B. Cross-module queries

With 11 briefs in one database, SQL queries answered questions that separate documents could not answer directly.

Deadlines that are close together



Missing assessment information



The missing-information queries also exposed gaps in the briefs themselves, showing that a central database can reveal inconsistencies in how briefs are written.

C. Chatbot

We wrote five test questions in four groups: facts, rules, meaning, and information not in the brief. We hoped to compare a baseline, which sends the whole brief to the model, with retrieval, which sends database facts plus the most relevant text sections. However, long running times and 503 errors limited our ability to collect results.

D. Costs & Security

Using a university-managed centralised repository and internally approved tools could offer benefits in terms of both cost and security. Rather than relying entirely on external services such as Gemini or ChatGPT, the university would have greater control over how assessment materials and student data are stored, accessed and managed. It could also reduce the risk of sensitive information being shared with third-party providers.

In terms of cost, developing and maintaining an internal system would require an initial investment in infrastructure, development and ongoing maintenance. However, it could reduce long-term reliance on external API subscriptions and usage-based pricing. A centralised repository could also reduce duplication by allowing different departments to access and reuse existing resources.

Although external tools may offer quicker implementation and access to advanced AI models, they introduce additional considerations around data protection, licensing, API costs and reliance on third-party services. An internally managed solution could provide greater institutional control and consistency, provided the university has the resources and expertise to maintain it.

V. Evaluation

A. What worked

Separating structured facts from free text meant that deadlines, weightings and criteria could be answered exactly from the database, with search needed only for long descriptions. Validating against a schema made gaps visible instead of hiding them, and keeping each record's source file, along with both the parsed date and the original wording, kept the data traceable. Caching, retries and a fallback model made batch ingestion practical despite frequent service overload errors.

B. Limitations

- Content that exists only as images was not captured.
- The evaluation was done by one person. It involved no students or lecturers.
- Brief text and student drafts were sent to an external AI service. A real deployment would need a

university-approved or in-house model, and until then this weakens the security argument for building in-house.

C. Research Questions

TABLE I. WHAT THE PROTOTYPE SHOWED

Question	***Evidence***	***Outcome***
Can assessment information be held in one structured database?	Section IV - A	Supported
Does centralisation give an overview across modules?	Section III -D	Supported
Can the database support tools for students and staff?	Section IV – C	Working prototype; not tested with users
Can the data reveal wider issues in briefs?	Section IV-B	Party: issues in briefs only
Can AI identify where students and lecturers struggle?	None	Not tested: needs question logs and performance data

These questions could be further explored if this project is taken onboard and expanded

VI. Conclusion

We investigated whether assessment information from separate documents could be centralised in one structured database, and what could be built on it. The database supported cross-module queries that scattered documents could not. A chatbot and a draft-feedback tool were built by reading from the database instead of reprocessing documents, showing that one repository can serve several applications.

The overarching limitation is the server being overwhelmed with requests, which slowed response times, the quality of the source briefs, and the use of an external AI service. Future work should extend the approach to more departments and document types, add module and performance data under proper governance to support analytics, test the tools with students and staff, and host the AI components within the university. Overall, the prototype suggests that a centralised assessment database is feasible and useful as a foundation, provided that these limitations are addressed.

Acknowledgment

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References

[1] E. A. Onyame, S. Annim, A.-G. Sayibu, D. S. Y. Amedahe, N. Borketey-Coffie, and H. N. Manu, 'Bridging Knowledge Silos at the University of Cape Coast: The Role of Cross-Departmental Knowledge Models in Enhancing Knowledge Sharing and Institutional Memory', Intelligent Information Management, vol. 17, no. 04, pp. 117–139, 2025, doi: 10.4236/iim.2025.174007.

[2] Y. Zhang, 'A Study of the Digital Transformation and Efficiency Improvement Path of University Administration', SHS Web of Conferences, vol. 213, p. 01001, 2025, doi: 10.1051/shsconf/202521301001.